

Theorem 1. Assumptions:

- *Let R be a ring.*
- *Let S be a ring.*
- *Let $f : R \rightarrow S$ be a ring homomorphism.*
- *Let $r \in R^\times$ be a unit.*

Conclusion: *Let $f(r) \in S^\times$ be a unit.*

Proof. Since f is a ring homomorphism, it maps units to units.

□